Voc4Cat: a sustainable, high TRL toolkit bridging cross-disciplinary applications in support of FAIR data-driven catalysis.

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Introduction

Creating shared, machine-readable vocabularies is essential for FAIR-compliant[1] (meta)data annotation, enabling machine-readability. In the field of catalysis, no such vocabulary existed which motivated the creation of Voc4Cat. Every concept in Voc4Cat is assigned a unique resolvable persistent identifier and reflects the community's shared understanding of the concept's definition. Voc4Cat, is a high technology readiness level (TRL 8) toolkit for the catalysis community to utilize and contribute to.

Voc4Cat integrated data flow

Figure 1 illustrates the complete data flow in Voc4Cat including some examples of current and future integrations in tools and services (more details are included in Table 1). A community member can access Voc4Cat either directly from its GitHub repository[2], or from its home page[3]. Editing the vocabulary (in .xlsx format), such as correcting definitions or suggesting new concepts, should follow the established guidelines[4,5]. The edited version of the vocabulary, through a pull request is processed by an automated continuous integration pipeline. Voc4Cat's curators then review the changes, providing feedback to the submitter (if needed), or directly approve them.

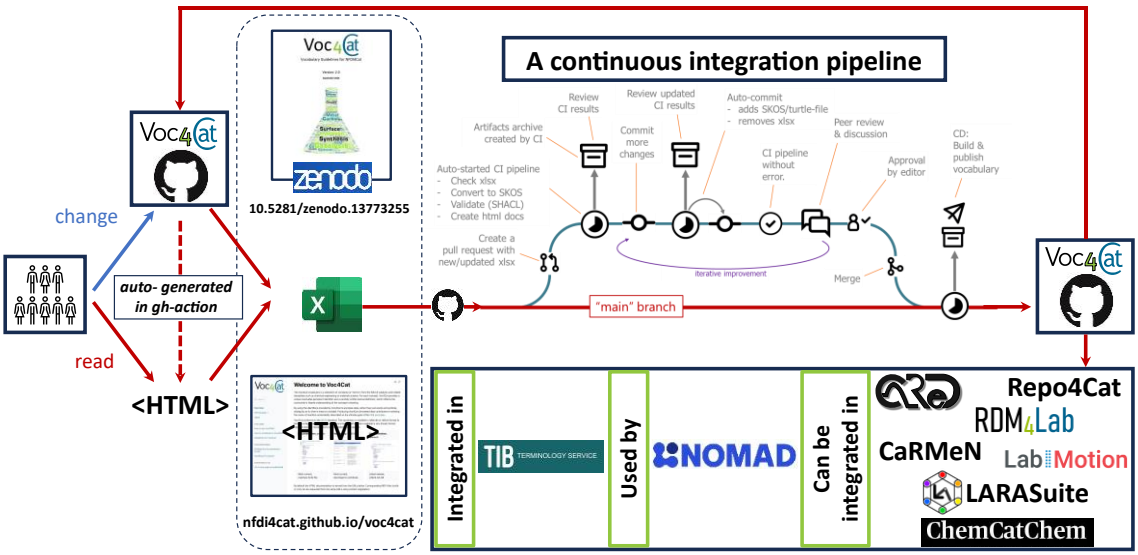

Figure 1.: Data flow in Voc4Cat and its current and potential future applications.

Voc4Cat used in various applications

Table 1 presents the main application categories for Voc4Cat, along with examples and potential benefits from its integration.

Table 1.: Application categories, examples, and Voc4Cat integration benefits.

Category	Examples of applications	Benefits from possible integration
Electronic lab notebooks (ELNs)	NOMAD ARE LARASuite LabMotion	**Enhanced data precision**: Voc4Cat standardizes terminology, ensuring precise, unambiguous descriptions of experimental conditions, catalysts, and reactions, which minimizes misinterpretations in data analysis and sharing.
Data repositories	Repo4Cat CaRMeN	**Enhanced data searchability**: Consistent, field-specific concepts make it easier to categorize and retrieve data within a repository, allowing researchers to find relevant datasets quickly and efficiently.
Education	RDM4Lab	**Enhanced reporting**: Use of consistent, standardized terms, facilitating faster evaluation of university reports. **Enhanced education**: Students get more familiar on FAIR (meta)data principles.
Scientific publishing	ChemCatChem The European Society Journal for Catalysis	**Enhanced visibility**: Use of semantically annotated articles and data that use standardized metadata.

Voc4Cat designed for sustainability

Voc4Cat is a sustainable, community-curated scientific resource guided by the recently established O3 principles[6] which stress the importance of:

1. Open data: Voc4Cat is CC0 licensed (documentation: CC-BY).
2. Open tools: All maintenance tools are developed on GitHub under an open license, including the voc4cat-tool[7], voc4cat-template[8], and a Python Package[9].
3. Open infrastructure: Hosting and development rely solely on existing open free services: a) GitHub for hosting code, documentation, and the vocabulary resource, and b) the w3id.org service for permanent URIs.

References and links

1. Wilkinson, M., et al., *Sci Data* 3, 160018 (2016). DOI: 10.1038/sdata.2016.18
2. https://github.com/nfdi4cat/voc4cat
3. https://nfdi4cat.github.io/voc4cat/
4. https://zenodo.org/records/13773255
5. https://nfdi4cat.github.io/voc4cat/docs_usage/published-guidelines-v2.html
6. Hoyt, C.T., et al., *Sci Data* 11, 547 (2024). DOI: 10.1038/s41597-024-03406-w
7. https://github.com/nfdi4cat/voc4cat-tool
8. https://github.com/nfdi4cat/voc4cat-template
9. https://pypi.org/project/voc4cat

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